SCIENCE & INNOVATION FOR SUSTAINABILITY

PROJECT REGISTRATION DRAFT & STANDALONE ABSTRACT

KRISHI-DRISHTI — Student Science Project Documentation & Submission Entry

Document Purpose: *This document contains our filled project registration draft and standalone project abstract. It is formatted cleanly so it can be adapted to any science exhibition, innovation congress, or academic competition. All project facts, costs, and experimental findings reflect actual physical bench testing and field observations.*

PART I: PROJECT REGISTRATION & ENTRY DETAILS

1. State:	Bihar	**District:**	[My District / Region Name]
2. Block / Taluka:	[My Local Block / Area]	**Area Type:**	Rural (Agricultural area) [X] Urban [ ]
3. Project Title:	**KRISHI-DRISHTI: An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture**		
4. Main Theme:	Science and Innovation for Sustainability	**Sub-Theme:**	Food, Agriculture and Health
5. Working Language:	English / Bilingual	**Target Group:**	Senior School Student Category (Class 9 / Age 14–15)
6. School / Institution:	[Full Official Name of Secondary School / Institution]		
7. Institution Address:	[Village / Town, Post Office, Block, District], Bihar — PIN: [_____]		
8. Contact Details:	Phone: [_____] | Email: [_____]		
9. Student 1 (Group Leader):	Name: [Full Name of Student 1] Class: Class 9 | Date of Birth: ____/____/2011 (Age: 14–15 Years) Contact Mobile: [_____]		
10. Student 2 (Co-Worker):	Name: [Full Name of Student 2 / Teammate] Class: Class 9 | Date of Birth: ____/____/2011 (Age: 14–15 Years) Contact Mobile: [_____]		
11. Guide Teacher / Mentor:	Name: [Guide Teacher Full Name] Designation: [Science Teacher / TGT / PGT] Contact Number: [_____]		

Student Signature	**Guide Teacher**	**Head of Institution / Principal**	**Event / Coordinator Verification**
_____ Date: ____/____/2026	_____ Date: ____/____/2026	_____ (Signature & Official Stamp)	_____ (Verification Seal / Date)

PART II: STANDALONE PROJECT ABSTRACT

(Concise summary formatted for exhibition booklets and project registries; exactly 284 words)

Project Name:	KRISHI-DRISHTI: An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture
Theme & Sub-Theme:	Science and Innovation for Sustainability — Food, Agriculture and Health
Central Research Question:	Can a low-cost, student-built handheld wand with an ESP32 microcontroller, a distance shroud, and an offline phone help farmers spot potato and tomato blights early right in the field without any internet?

Last year, agricultural news and social media showed devastating crop failures across major tomato and potato belts. In tomato hubs like Kolar and Nashik, early blight and fruit rot wiped out harvests, forcing farmers to dump damaged crates on roads. Just months later, winter fog hit northern India, and potato farmers across Samastipur, Vaishali, and Agra saw their fields collapse into black, rotting slime from Late Blight within 72 hours. Seeing the exact same tragedy happen across places A, B, C, and D, I investigated why farmers could not catch the infection early. In online farmer groups, growers desperately asked what chemicals to spray, but by then spraying was useless. Existing mobile apps fail because 4G cellular signals drop to zero out in open fields, and uncalibrated phone cameras take blurry photos blinded by sun glare.

To solve this, I designed KRISHI-DRISHTI, an ultra-low-cost (under Rs 1,500) offline screening kit. It pairs an ESP32 controller wand (with an OLED screen and a DHT11 canopy humidity sensor) with an ESP32-CAM unit fitted with an 8 cm black optical hood that standardizes focal distance and blocks harsh sunlight. The camera creates an offline Wi-Fi hotspot, sending images to an old Android phone in the pocket that runs a quantized MobileNetV3 AI model in just 4.9 ms without internet.

On 60 clean botanical test leaves, the system achieved 88.33% exact diagnosis (53/60) and 90.0% pathogen-family correlation. On 93 messy outdoor PlantDoc photos, raw accuracy was 33.33%, but dual-stage Shannon entropy gating safely rejected 37 confusing images as UNCERTAIN, raising accepted accuracy to 55.36%. Physical testing revealed real limitations: an unseen tea leaf reached 87% false confidence, and night flashlight shadows mimicked blight. KRISHI-DRISHTI provides an honest, accessible preliminary screening tool that aids farmers in early organic intervention.

Keywords: *Offline Edge-AI, ESP32 Wand, Late Blight, Tomato, Potato, Optical Shroud, Smallholder Farmers.*

PART III: PROJECT CHECKLIST & QUICK REVIEW GUIDE

Key requirements to review before exhibition display or project submission:

#	Project Item / Requirement	Current Readiness Status	Status
1	Project Registration / Cover Details Filled	Draft completed on Page 1; ready for school signatures	[X] Ready
2	Main Project Report within word limits	Full 32-section report compiled (~2,950 body words)	[X] Done
3	Research Logbook / Daily Work Diary	Template & daily entries compiled in Document 02	[X] Ready
4	Technical Blueprints & Schematic Annexure	Pinouts, ASCII schematic, and math compiled in Document 03	[X] Ready
5	Presentation Posters & Oral Defense Scripts	All four poster blueprints with exact angles in Document 05	[X] Ready
6	Physical Hardware Prototype Functional	Controller wand, camera shroud, and pocket phone tested	[X] Working

Quick Facts for Judges & Reviewers:

Evaluation Parameter	Exact Technical Observation / Project Value
Total Hardware Bill of Materials	Rs 1,470 (ESP32 Rs 420, ESP32-CAM Rs 540, OLED Rs 190, DHT11 Rs 95, SD Card Rs 160, Shroud Rs 35, Cap/Wires Rs 30)
Offline Latency & Connectivity	4.94 ms reference inference; 12–15 ms on budget phone; 100% offline local Wi-Fi mesh (192.168.4.1); zero cellular data needed.
Dataset Cryptographic Audit	7,710 records reconciled into 6,965 unique image hashes; 5 duplicate pairs audited (zero conflicting labels); 100% zero data leakage.
Botanical Benchmark Accuracy	60 fresh leaves: 88.33% strict (53/60), 92.98% accepted (53/57, 3 gated uncertain), 90.0% pathogen-family correlation (54/60).
Outdoor PlantDoc Stress Test	93 unconstrained field photos: 33.33% raw baseline (31/93) -> 55.36% accepted (31/56) after Shannon entropy gating 37 ambiguous frames.
Honest Scientific Limitations	Tea leaf reached 87% false confidence (closed-set limitation); steep night torch shadows triggered 97.9% false early blight prediction.